CubeSat Communications Board
70 cm BPSK TX / 2m AFSK RX

Project: CentiSat CubeSat -- Senior Capstone Design

Board: Communications Subsystem, Single PCB

Designer: Nick Grabbs / KI5YNG

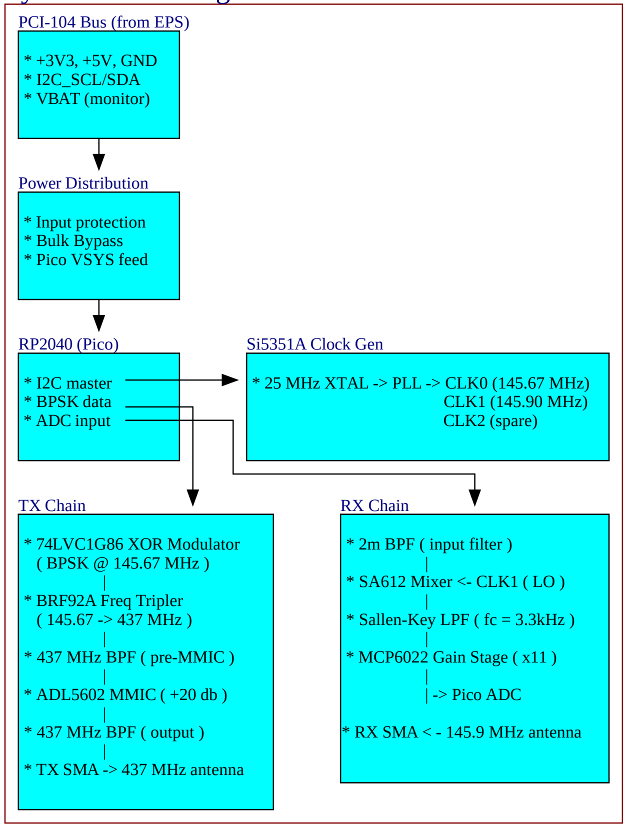
Revsion: 0.2
Date: 03/13/2026

Repsoitory: centisat/hardware/comms

Acronym Legend

* EPS - Electrical Power System

System Block Diagram



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Power Rail Summary

Power Rails

Rail	Source	Voltage	Consumers
+3V3	EPS via PCI-104 (a1, a2)	3.3V ±2%	RP2040, Si5351A, 74LVC1G86, MCP6022
+5V	EPS via PCI-104 (a4, a5)	5.0V ±2%	SA612, ADL5602, tripler bias, Pico VSYS
GND	EPS via PCI-104 (a3, a6, a8, a12)	0V	All
VBAT	EPS via PCI-104 (a7)	6.0–8.4V	Monitor only (not used for power)

Estimated current budget: +3V3: ~150 mA (RP2040 ~50mA, Si5351A ~30mA, logic ~10mA, op-amp ~10mA) +5V: ~80 mA (SA612 ~3mA, ADL5602 ~60mA, tripler ~5mA, Pico VSYS ~10mA) Total: <250 mA — well within EPS TPS62933F capacity (3A per rail)

Net Name Convention

Power rails use **power port symbols** (global across all sheets). Signal nets use **net labels** (also global in flat design).

Net	Type	Description
+3V3, +5V, GND, VBAT	Power port	Supply rails (global)
I2C_SDA, I2C_SCL	Net label	I2C bus (Si5351A + FC bus)
CLK0_OUT, CLK1_OUT	Net label	Si5351A clock outputs
BPSK_DATA	Net label	RP2040 → XOR modulator

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Net	Type	Description
TX_OUT, RX_IN	Net label	RF paths to SMA connectors
RX_BASEBAND	Net label	MCP6022 output → Pico ADC

Key Design Notes

1. Power from EPS via PCI-104 bus — no on-board regulators
2. BPSK modulation at 145.67 MHz (before tripler) — 74LVC1G86 fails at 437 MHz, see overview.md timing analysis
3. Odd-order frequency multiplication preserves BPSK phase states ($3 \times 180° = 540° \equiv 180°$ mod 360°)
4. Tripler bias 47kΩ — empirically optimized (-7.4 dBm @ 437 MHz)
5. All RF filter caps must be C0G/NP0 (not X7R)
6. I2C addresses: Si5351A = 0x60, EPS LTC4162 = 0x68 (no conflict)
7. Flat schematic design — nets connect by name across sheets
8. Prototype uses Pico module; bare RP2040 deferred to v2

CLOCK GENERATOR

CLOCK GENERATOR -- Si5351A (I2C address 0x60). Generates three independent clock outputs from a single 25 MHz crystal. CLK0 = 145.67 MHz (TX carrier, tripled to 437 MHz). CLK1 = 145.9 MHz (RX local oscillator for SA612 mixer). CLK2 = spare. All outputs driven at 8 mA (max) intio 50Ohm loads. Controlled by RP2040 over I2C.

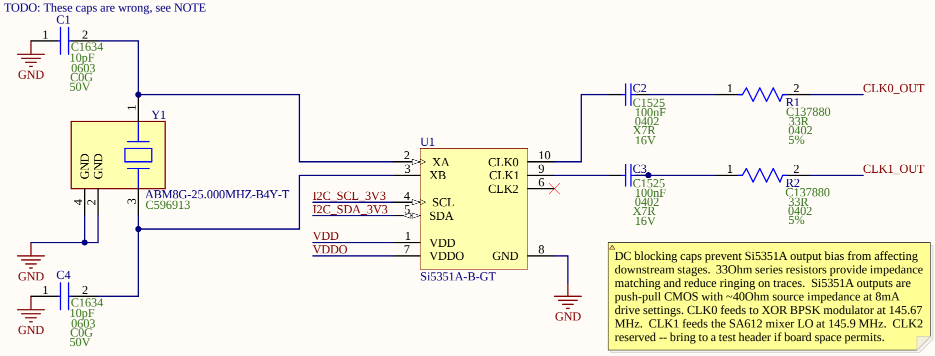
Si5351A IC

Crystal load capacitance:

CL = (Not found x Not found)/(Not found + Not found) + Cstray.

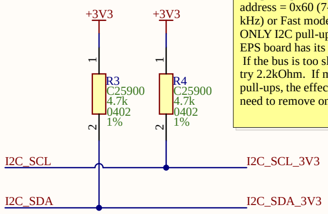
With Not found = Not found = 10pF and ~3pF stray: CL ~8pF.

Match to crystal's specified load capacitance. If crystal specifies 8pF load, use 10pF caps. If crystal specifies 10pF load, use 15pF caps. C0G/NP0



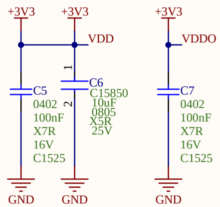
DC blocking caps prevent Si5351A output bias from affecting downstream stages. 33Ohm series resistors provide impedance matching and reduce ringing on traces. Si5351A outputs are push-pull CMOS with ~40Ohm source impedance at 8mA drive settings. CLK0 feeds to XOR BPSK modulator at 145.67 MHz. CLK1 feeds the SA612 mixer LO at 145.9 MHz. CLK2 reserved -- bring to a test header if board space permits.

I2C Pull-Up Resistors



I2C pull-ups: 4.7kOhm to +3V3. Si5351A address = 0x60 (7-bit). Standard mode (100 kHz) or Fast mode (400 kHz). These are the ONLY I2C pull-ups on the comms board -- the EPS board has its own pull-ups for the LTC4162. If the bus is too slow (rounded edges on SCL), try 2.2kOhm. If multiple devices on the bus have pull-ups, the effective resisance drops --- may need to remove one set.

Bypass Capacitors



Not found / Not found at VDD: place Not found (100nF) directly at pin 1 with shortest possible traces. Not found (10uF) can be further away.

Title: CubeSat Comms -- Clock Generator		
Size: A3	Number	Revision: A
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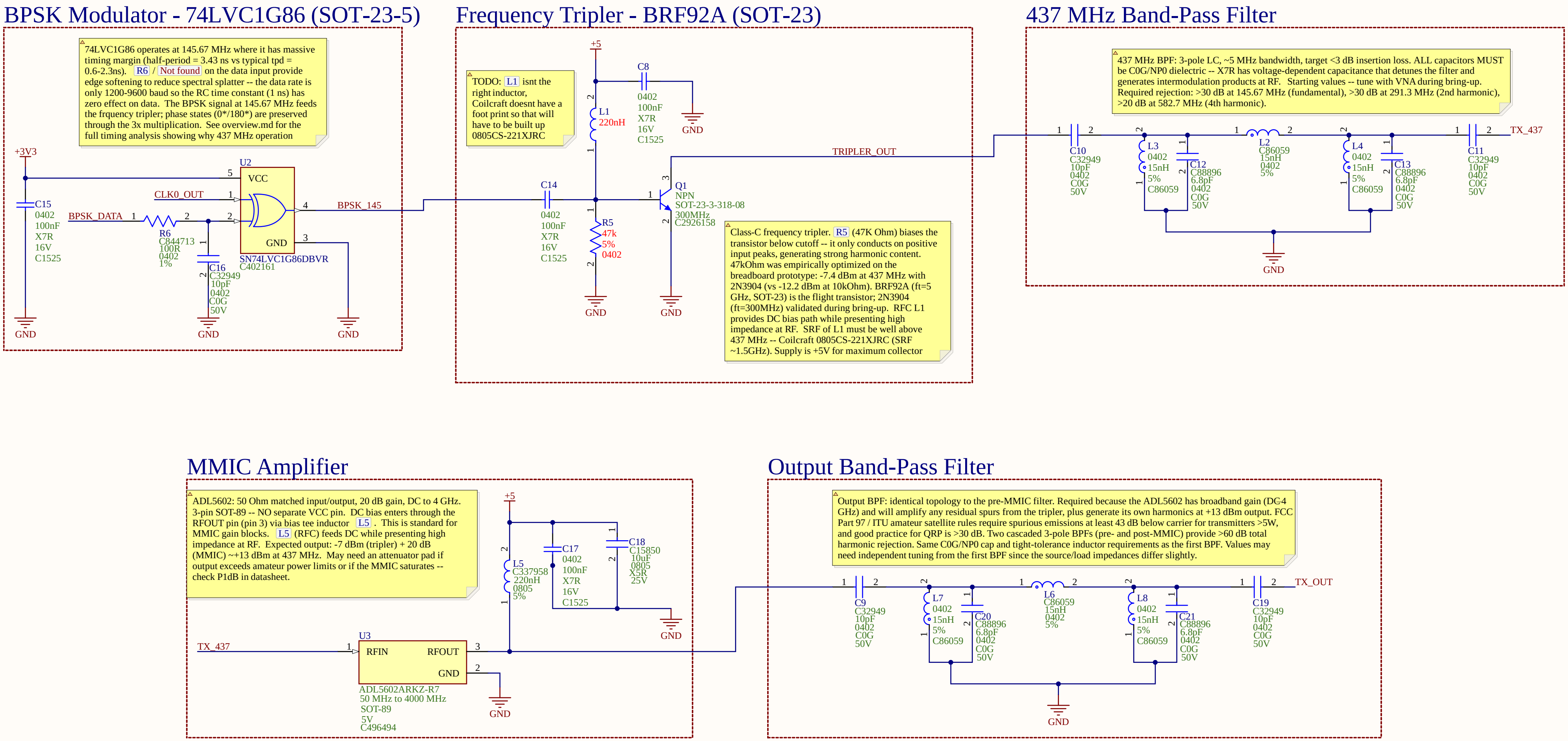
File: C:\Users\..\Clock Gen.SchDoc		Drawn By: Nick Grabbs - KI5YNG

TX CHAIN

TX CHAIN - 437 MHz BPSK downlink transmitter. The carrier from the Si5351A (145.67 MHz) is BPSK-modulated by the 74LVC1G86 XOR gate, then frequency-tripled to 437 MHz by a class-C biased BFR92A transistor. A 3-pole LC BPF selects the 3rd harmonic and rejects the fundamental and 2nd harmonic. The ADL5602 MMIC provides ~20 dB gain to bring the signal to ~+10 dBm output. BPSK phase states are preserved through odd-order multiplication: 3 x 180* = 540* = 180*



Title: CubeSat Comms -- TX Chain		
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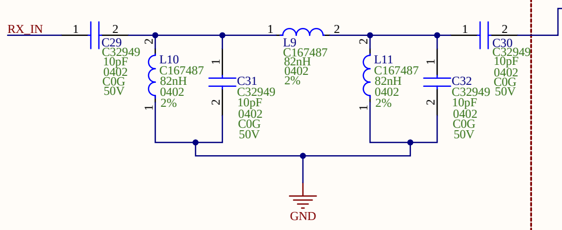
RX CHAIN

RX CHAIN — 145.9 MHz FM AFSK uplink receiver. A 2m band-pass filter at the input rejects out-of-band signals before the mixer. The SA612 double-balanced mixer downconverts the 2m signal to baseband using the Si5351A CLK1 as LO. A 2nd-order Sallen-Key LPF (fc ≈ 3.3 kHz) filters the baseband, passing the 1200/2200 Hz AFSK tones while rejecting mixer products and noise. The MCP6022 U7B stage provides ~20 dB gain before the RP2040 ADC. Single-supply operation with 1.65V virtual ground (mid-supply bias).

2m Input Band-Pass Filter

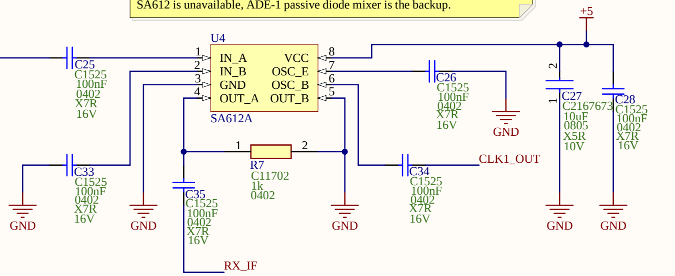
2m input BPF: 3-pole LC, centered ~146 MHz, ~4 MHz bandwidth. Critical for protecting the SA612 mixer from overload. Must reject:

* 437 MHz TX leakage (>40 dB — this is your own transmitter)
* FM broadcast 88–108 MHz (>30 dB — these signals are very strong)
* 2nd harmonic of LO at 291.8 MHz (>20 dB) ALL capacitors must be C0G/NP0. Starting values are estimates — tune L and C values with VNA to center the passband on 145.9 MHz and achieve <3 dB insertion loss at the operating frequency. If insertion loss is too high, consider a 2-pole filter instead (remove the center L10 coupling and one shunt section).



SA612 Mixer

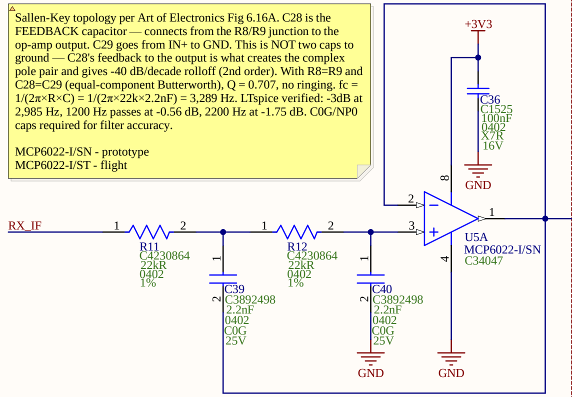
SA612A: Gilbert cell mixer, suitable for VHF mixing down to baseband. Internal oscillator not used — external LO from Si5351A CLK1 injected on pin 6. Single-ended RF input (pin 2 AC grounded). R7 (1kΩ) between pins 4/5 sets the IF output impedance and provides a DC path. IF output is differential but we take single- ended from pin 4 through C25 to the baseband filter. VCC = +5V (not +3V3 — SA612 min supply is 4.5V). Current consumption ~3mA. If SA612 is unavailable, ADE-1 passive diode mixer is the backup.



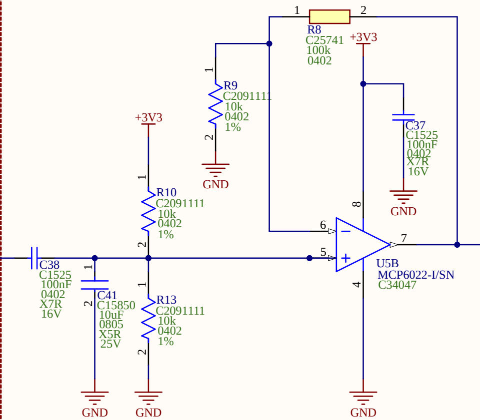
Sallen-Key Low-Pass Filter

Sallen-Key topology per Art of Electronics Fig 6.16A. C28 is the FEEDBACK capacitor — connects from the R8/R9 junction to the op-amp output. C29 goes from IN+ to GND. This is NOT two caps to ground — C28's feedback to the output is what creates the complex pole pair and gives -40 dB/decade rolloff (2nd order). With R8=R9 and C28=C29 (equal-component Butterworth), Q = 0.707, no ringing. fc = 1/(2π×R×C) = 1/(2π×22k×2.2nF) = 3,289 Hz. LTspice verified: -3dB at 2,985 Hz, 1200 Hz passes at -0.56 dB, 2200 Hz at -1.75 dB. C0G/NP0 caps required for filter accuracy.

MCP6022-I/SN - prototype
MCP6022-I/ST - flight



Baseband Gain Stage



Non-inverting amplifier, gain = 1 + 100k/10k = 11 (20.8 dB). R10/R11 bias the input to 1.65V (mid-supply) for single-supply operation. C32 (10µF) decouples the bias point at AC so it acts as a solid virtual ground for the signal. C30 AC-couples the Sallen-Key output to remove any DC offset from the filter. Output goes to RP2040 ADC (GP26/ADC0). Full-scale ADC input is 0–3.3V; with 1.65V bias, the signal can swing ±1.65V before clipping. Gain of 11 may need adjustment based on actual signal levels — change R13 to tune. MCP6022 GBW = 10 MHz, well above the ~3 kHz signal band.

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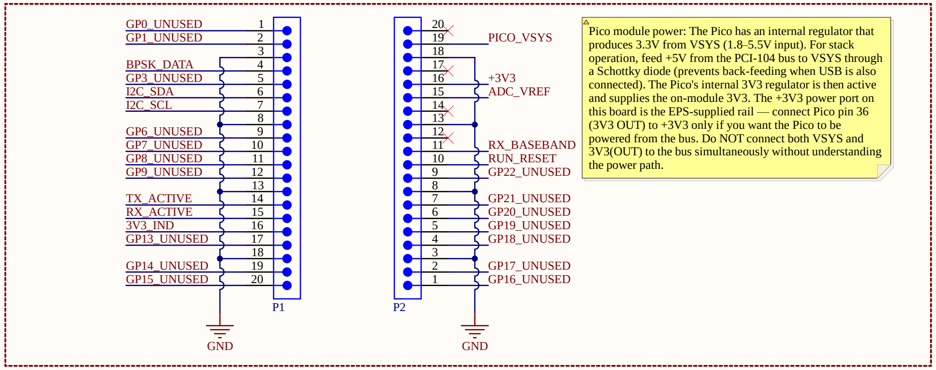
File: C:\Users\..\RX Chain.SchDoc		Drawn By: Nick Grabbs - KI5YNG	

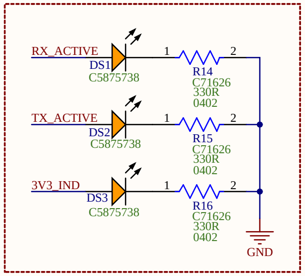
DIGITAL CONTROL

DIGITAL CONTROL — Raspberry Pi Pico module (RP2040-based). Handles BPSK/DBPSK encoding for TX, AFSK demodulation for RX, Si5351A configuration via I2C, and command processing. Using Pico module for prototype — all RP2040 support circuitry (flash, regulator, crystal) is on-module. Bare RP2040 redesign deferred to v2 flight board after RF chain validation.

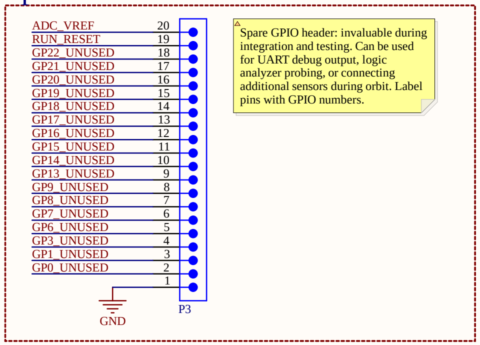
Pico Module



Status LED's



Spare GPIO Header



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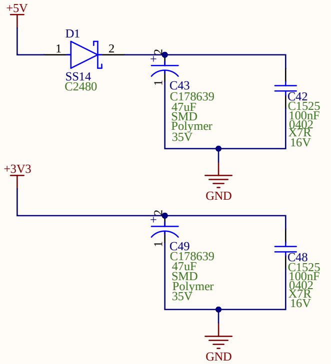
POWER DISTRIBUTION

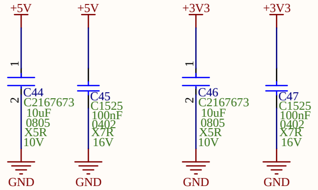
POWER DISTRIBUTION — The comms board receives +3V3 and +5V from the EPS via the PCI-104 stack connector. No on-board voltage regulation. This sheet provides input protection, bulk bypass capacitors, and distributes power to all subsystems. Power budget: ~150 mA on +3V3 (RP2040 + Si5351A + logic), ~80 mA on +5V (SA612 + ADL5602 + tripler bias). Total: <250 mA — well within EPS TPS62933F capacity (3A per rail).

Input Protection and Filtering



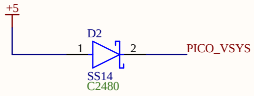
D4 provides reverse polarity protection on the +5V rail. Adds ~0.3V forward drop — the ADL5602 and SA612 can tolerate this (both specified for supply down to 4.5V, so 5.0-0.3=4.7V is fine). The +3V3 rail does NOT have a series diode because the RP2040 and Si5351A are sensitive to voltage drop — 3.0V may be too low for reliable operation. If reverse polarity protection is needed on +3V3, use a P-MOSFET instead of a Schottky (near-zero drop when conducting). Bulk caps dampen transients from the PCI-104 connector traces. The EPS has its own bulk caps at the connector — these are the receiving-end complements.

Per-Subsystem Bypass



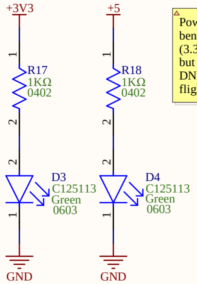
Board-level bulk bypass at the power distribution star point. Local bypass at each consumer IC is the responsibility of that IC's sheet (C1/C2/C3 on Clock Gen, C8 on TX, C21/C22 on RX, etc.). These caps cover trace inductance between the PCI-104 connector and the far side of the board. Place Not found / Not found near the center of their respective power distribution traces.

Pico Module Power Feed



D5 isolates the stack +5V from the Pico's USB VBUS. When USB is connected, the Pico's internal diode on VBUS feeds VSYS. When running from the stack (no USB), D2 feeds VSYS from +5V. The Pico's internal 3.3V buck-boost regulator produces 3V3 from VSYS. Drop across D5 (~0.3V) gives VSYS ≈ 4.7V — well within the Pico's VSYS range (1.8–5.5V).

Power Indicator LEDs



Power indicator LEDs for bench debugging. I = (3.3-2.0)/1k = 1.3mA (dim but visible). Mark R17/R18 as DNP-optional in BOM for flight.

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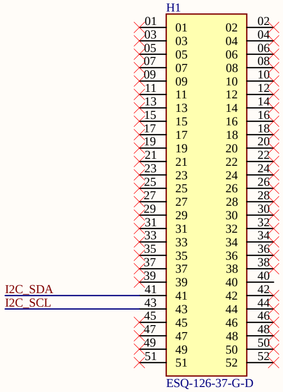
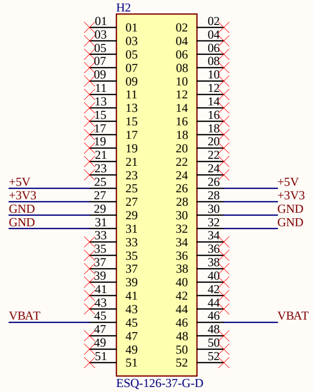
Connectors

CONNECTORS — CSKB H1 + H2 stack connectors for inter-board power and signal distribution, SMA connectors for TX and RX antenna ports, and test points for bench validation. All power (+3V3, +5V) arrives on H2; I2C, SPI, and COMMS_IRQ route through H1. The comms board is a power CONSUMER — it receives regulated rails from the EPS.

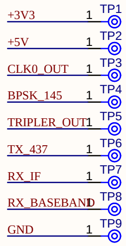
CSKB H1 + H2 Stack Connectors



CSKB stack connectors: 2× Samtec ESQ-126-39-G-D (2×26, 0.1″ (2.54 mm), stackthrough headers, one each for H1 and H2). Pin assignment locked to the Pumpkin CSKB H1/H2 convention — all boards in the centisat stack (EPS, FC, comms, payload) use the same H1/H2 pin numbers for the same signal classes. The comms board is a power CONSUMER on all power pins and a SPI SLAVE to the FC. COMMS_IRQ (H1.16) is driven by the comms RP2040 as a data-ready interrupt to the FC. I2C is shared between EPS LTC4162 (0x68) and comms Si5351A (0x60) — no address conflict. See system/interfaces/cskb_pinmap.md for the full canonical pin map.

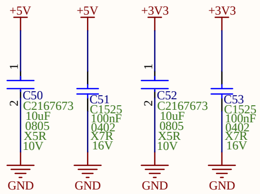


Test Points



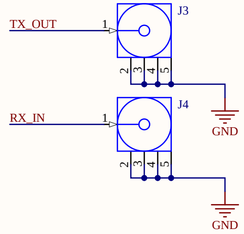
Test points are critical for bring-up and debugging. TP3 - TP6 trace the TX signal through the chain — essential for validating each stage independently. TP7 - TP8 trace the RX chain. Use short probe leads on RF test points to minimize added inductance. TP9 (GND) should be a loop-style test point for easy scope ground clip attachment. For flight boards, reduce to TP1 / TP2 (power) + TP9 (GND) minimum.

Bulk Bypass at Stack Connector



Connector-side bypass caps are the receiving complement to the EPS board's connector-side caps (C60–C63 on EPS Sheet 3). Together they form a low-impedance distributed bypass across the stack connector. Place these caps as close to the connector pins as possible — trace inductance between connector and cap defeats the purpose.

RF Connectors



Separate TX and RX SMA connectors — no diplexer for prototype. For flight, a diplexer can be added between a single antenna and the two SMA ports, or external to the board. SMA center pin impedance is 50Ω — match traces accordingly during PCB layout. Keep SMA-to-first-component traces as short as possible, with continuous ground plane underneath. Controlled-impedance traces (50Ω microstrip or CPWG) required for RF paths.

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File: C:\Users\..\Connectors.SchDoc		Drawn By: Nick Grabbs - KI5YNG

Name	Description	Designator	Footprint	LibRef	Quantity
CL10C100JB8NNNC	Capacitor	C1, C4	CAPC1608X90N	CL10C100JB8NNNC	2
CL05B104KO5NNNC	Chip Capacitor, 100 nF, +/- 10%, 16 V, -55 to 125 degC, 0402 (1005 Metric), RoHS, Tape and Reel	C2, C3, C5, C7, C8, C14, C15, C17, C25, C26, C28, C33, C34, C35, C36, C37, C38, C42, C45, C47, C48, C51, C53	CAPC1050X55X23ML05T13	CMP-2000-05439-1	23
CL21A106KAYNNNE	CL21 Series 0805 10uF 25V ±10% Tolerance X5R Multilayer Ceramic Chip Capacitor	C6, C18, C41	FP-CL21-IPC_A	CMP-13271-002975-1	3
CL05C100JB5NNNC	Capacitor	C9, C10, C11, C16, C19, C29, C30, C31, C32	CAPC1005X55N	CL05C100JB5NNNC	9
GJM1555C1H6R8BB01D	Multi-Layer Ceramic Capacitor 6.8pF C0G ±0.1pF 0402 Paper T/R	C12, C13, C20, C21	FP-GJM155-0_05-IPC_A	CMP-2008-03772-2	4
C0805C106K8PACTU	CAP CER 10UF 10V X5R 0805	C27, C44, C46, C50, C52	FP-C0805C-DG-MFG	CMP-2007-03243-2	5
C0402C222F3GECAUTO	Capacitor	C39, C40	C0402	C0402C222F3GECAUTO	2
EEHZA1V470P	Aluminum Hybrid Polymer Capacitors 47uF 20% 35V Life 10000Hours AEC-Q200 RADIAL SMT	C43, C49	FP-EEH-D-MFG	CMP-05428-000653-1	2
SS14	DIODE SCHOTTKY 40V 1A SMA	D1, D2	FP-403AE-MFG	CMP-2000-04943-2	2
LTST-S270KGKT	LED GREEN CLEAR CHIP SMD R/A	D3, D4	FP-LTST-S270KGKT-MFG	CMP-2000-06629-2	2
APG1608SEKC/T	LED, SMT, 0603(1608), 0.25mm Tthickness, Super Bright Orange	DS1, DS2, DS3	KING-LED0603-25-ORANGE_V	CMP-0239-00002-1	3
ESQ-126-37-G-D		H1, H2	SAMTEC_ESQ-126-37-G-D	CMP-015-000000-1	2
60312242114510	Rf Coax Conn, Sma Jack, 50 Ohm, Pcb Rohs Compliant: Yes	J3, J4	60312242114510	CMP-03913-97004434-3	2
HLP2525CZERR20M01	Commercial Inductors, High Saturation Series 0.2uH 24A 3mΩ 20%	L1	FP-IHLP-2525CZ-FP_2_413x3-MFG	CMP-02424-000044-1	1
LQG15HS15NJ02D	Multilayer type RF Inductor 15nH ±5% 0.32Ω 450mA 0402	L2, L3, L4, L6, L7, L8	FP-LQG15HH_02-IPC_B	CMP-2100-03652-2	6
LQW2BASR22J00L	Wire Wound RF Inductor 220nH ±5% 400mA 0.7Ω 0805 (2015)	L5	FP-LQW2BAN_00-MFG	CMP-06042-011308-1	1
LQW15AN82NG00D	Wire Wound RF Inductor 82nH ±2% 130mA 2.24Ω 0402 (1005)	L9, L10, L11	FP-LQW15AN_00-W0_5-0_1-MFG	CMP-06042-015778-1	3
MMT-120-01-L-SH	Horizontal Surface Mount Header, -55 to 125 degC, 2 mm Pitch, 20-Pin, Male, RoHS	P1, P2, P3	SMTC-MMT-120-01-X-SH	CMP-1029-00077-1	3
MMBT3904LT1G	General Purpose Transistor, NPN Silicon, 3-Pin SOT-23, Pb-Free, Tape and Reel	Q1	ONSC-SOT-23-3-318-08_V	CMP-1048-00631-1	1
RC0402JR-0733RL	RES SMD 33 OHM 5% 1/16W 0402	R1, R2	FP-RC0402-0_4-MFG	CMP-2002-07987-2	2
0402WGF4701TCE	Resistor	R3, R4	RESC1005X40N	0402WGF4701TCE	2
MCR004QLPJ473	Thick Film Chip Resistor, 01005, 47kΩ, 5%, 250ppm/°C, 0.031W, 15V	R5	FP-MCR004-IPC_C	CMP-08839-015286-1	1
	RES 100 OHM 1%		FP-CRCW0402-C-e3-		